

Estimating Social Interactions with Peer-of-Peers Instruments

A solution to the assignment (Campa & De Giorgi, Uniandes 2026)

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Summary. We relax the class exclusion restriction by letting *direct peers' characteristics* affect a student's outcome directly. This makes direct-peer instruments invalid and the class estimator inconsistent. We identify the model instead with *peers-of-peers* characteristics (G^2x) as excluded instruments, estimated by the same concentrated two-step GMM with the CES nonlinearity. On synthetic data with heterogeneous schools, the class estimator badly overstates the peer effect ($\hat{\lambda} = 0.92$ vs. true 0.30, $\hat{\beta} = 1.97$ vs. 5.0), while the peer-of-peers estimator recovers every parameter.

1. Model and the exclusion violation

Let y_i be GPA and g_{ij} the (row-normalized) weight i puts on peer j . The CES peer norm is $\tilde{y}_i(\beta, y) = (\sum_j g_{ij} y_j^\beta)^{1/\beta}$. The **extended** outcome equation adds a contextual term in direct peers' characteristics $(Gx)_i = \sum_j g_{ij} x_j$:

$$\text{connected: } y_i = x_i^\top \gamma + \underbrace{(Gx)_i^\top \phi}_{\text{contextual [new]}} + \lambda \tilde{y}_i(\beta, y) + \varepsilon_i; \quad \text{isolated: } y_i = x_i^\top \gamma + \varepsilon_i. \quad (1)$$

(We set $\delta = 1$ so own characteristics enter equally for both groups; this keeps the algebra light and preserves the identification problem.)

Item 1 — why the class estimator is inconsistent. In class, $\tilde{y}_i$ was instrumented with functions of direct peers' characteristics Gx , valid because Gx shifted peers' outcomes but was *excluded* from i 's own equation. Once $\phi \neq 0$, Gx enters i 's equation directly. An estimator that (a) omits the term $(Gx)^\top \phi$ and (b) uses Gx -based instruments therefore violates the exclusion restriction:

$$\mathbb{E}[z_i \varepsilon_i^{\text{class}}] = \mathbb{E}[z_i (\varepsilon_i + (Gx)_i^\top \phi)] \neq 0, \quad (2)$$

because z_i is built from the same Gx now sitting in the (composite) error. The direct-peer instrument is correlated with the omitted contextual term, so $\hat{\lambda}$ and $\hat{\beta}$ are inconsistent (Section 5 shows $\hat{\lambda}$ inflated to 0.92).

2. Identification with peers of peers

Item 2 — relevance. Let $(G^2x)_i = \sum_j g_{ij} (Gx)_j$ be the characteristics of i 's peers-of-peers. They shift the outcomes y_j of i 's direct peers (through *their* private and contextual terms), and hence move i 's norm $\tilde{y}_i$. Thus $\text{Cov}((G^2x)_i, \tilde{y}_i) \neq 0$: peers-of-peers' characteristics are relevant for the endogenous norm. (Empirically the first-stage correlation between the resulting predicted norm and the true norm is high; see the code's first-stage figure.)

Item 3 — validity. $(G^2x)_i$ are *valid excluded* instruments under the assumption that the direct effect of others' characteristics reaches only **direct** peers (distance 1). Formally,

$$\mathbb{E}[(G^2x)_i \varepsilon_i] = 0, \quad (3)$$

which holds if (i) only direct peers' characteristics enter i 's equation (no distance-2 contextual effect), and (ii) the network and errors are such that distance-2 characteristics are uncorrelated with i 's idiosyncratic shock (guaranteed by construction in our DGP: characteristics and errors are independent, with no homophily).

Item 4 — the identifying assumption, illustrated. Figure 1 shows a focal student i , the *direct peers* (distance 1) whose characteristics now enter i 's equation, and the *peers of peers* (distance 2) whose characteristics do not. In an intransitive triad $A \rightarrow B \rightarrow C$ with $A \not\rightarrow C$, x_C affects y_B (hence A 's norm) but is excluded from A 's equation, so x_C instruments A 's norm.

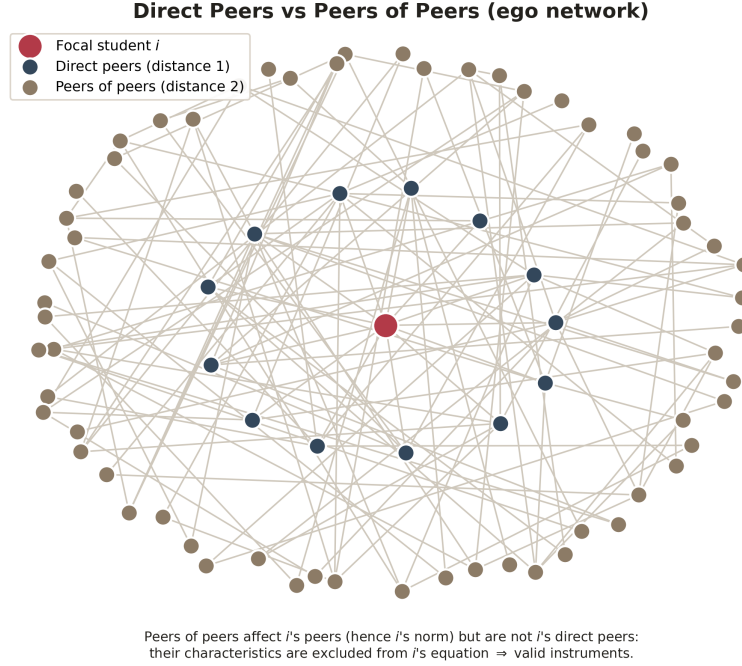


Figure 1. Direct peers (distance 1, in i 's equation) vs. peers of peers (distance 2, excluded $\Rightarrow$ instruments).

3. Estimation algorithm (Item 5)

We use the *same* concentrated two-step GMM as in class, changing only the controls and the excluded instruments.

1. **Within transform** by (school $\times$ isolation) to remove the intercept and school fixed effects.
2. **First stage / instrument construction.** Fit $\hat{y} = \text{OLS of } y \text{ on } [1, x, Gx, G^2x, \text{school dummies}]$. The key addition is G^2x , the peer-of-peers characteristics. Build the predicted norm and its curvature instrument

$$\hat{S}_i = \left(\sum_j g_{ij} \hat{y}_j^\beta \right)^{1/\beta}, \quad \hat{D}_i = \frac{\partial \hat{S}_i}{\partial \beta} = \hat{S}_i \left[\frac{1}{\beta} \frac{\sum_j g_{ij} \hat{y}_j^\beta \ln \hat{y}_j}{\sum_j g_{ij} \hat{y}_j^\beta} - \frac{1}{\beta^2} \ln \sum_j g_{ij} \hat{y}_j^\beta \right].$$

Because x and Gx are controls in the structural equation, the *excluded* identifying variation in $\hat{S}, \hat{D}$ is exactly the G^2x part of $\hat{y}$.

3. **Moments.** Isolated: $\mathbb{E}[x_i (y_i - x_i^\top \gamma)] = 0$. Connected: $\mathbb{E}[Z_i (y_i - x_i^\top \gamma - (Gx)_i^\top \phi - \lambda S_i(\beta))] = 0$ with $Z_i = (x_i, (Gx)_i, \hat{S}_i, \hat{D}_i)$. This is over-identified (11 moments, 8 parameters).
4. **Concentration + search.** Given β , the residual is linear in (γ, ϕ, λ) , solved analytically; we then minimize the GMM objective over the single nonlinear parameter β (grid + bounded search).

5. **Two-step and SE.** Identity weights first, then the optimal weight $\widehat{W} = \widehat{\Omega}^{-1}$, and school-cluster-robust standard errors (sandwich with the $G/(G-1)$ correction).

The **naive (class) estimator** we compare against omits the $(Gx)^\top \phi$ control and builds $\widehat{S}, \widehat{D}$ from a first stage on own covariates only $[1, x, \text{dummies}]$, so its instruments carry direct-peer (Gx) variation — invalid here.

4. Data

The synthetic environment matches the class DGP with two realistic changes. **(i) Heterogeneous schools:** each of the 150 schools draws its own size in $[80, 320]$ (here 30,261 students, mean size 202, 18% isolated), instead of identical schools. **(ii) Contextual effect:** direct peers' characteristics enter the outcome with $\phi = (0, 0.15, 0.30)$ on (age, female, parent-college). Realized GPA is generated from the fixed point $y = p + (Gx)\phi + \lambda S(y, \beta)$ and capped to $[1, 4]$ ($< 0.01\%$ clipped). True values: $\gamma = (0.10, 0.24, 0.40)$, $\lambda = 0.30$, $\beta = 5$.

5. Results (Item 6)

Table 1. True vs. naive (class, inconsistent) vs. peer-of-peers (correct). Cluster-robust SE for the correct estimator in brackets.

Parameter	True	Naive (class)	Peer-of-peers
γ_{age}	0.10	0.089	0.089 [0.001]
γ_{female}	0.24	0.214	0.215 [0.003]
$\gamma_{\text{parent coll.}}$	0.40	0.374	0.375 [0.004]
ϕ_{female}	0.15	— (omitted)	0.150 [0.004]
$\phi_{\text{parent coll.}}$	0.30	— (omitted)	0.303 [0.005]
λ (peer effect)	0.30	0.924	0.298 [0.005]
β (CES curvature)	5.00	1.97	4.98 [0.248]

The class estimator triples the peer effect (0.92 vs. 0.30) and badly misses the curvature (1.97 vs. 5), because the omitted contextual term loads onto the (invalid) direct-peer instruments. The peer-of-peers estimator recovers every parameter, including the contextual coefficients ϕ .

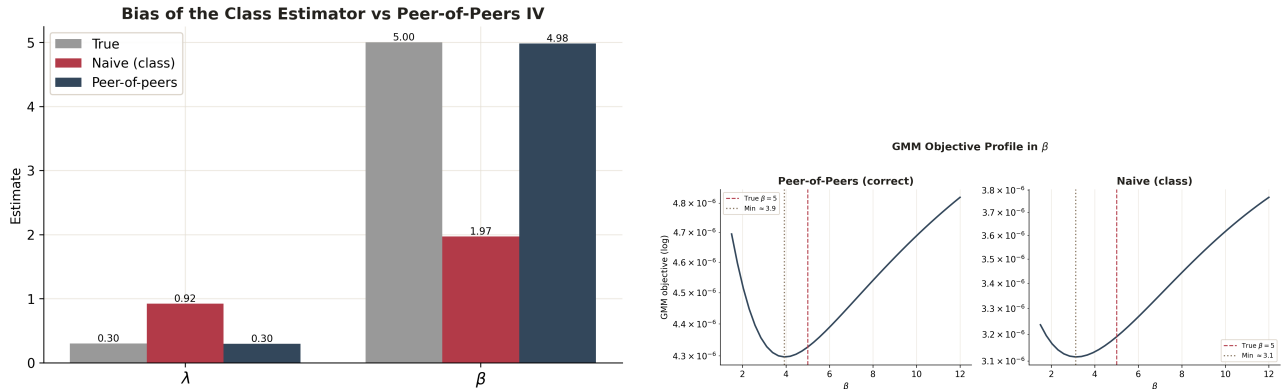


Figure 2. Left: bias of the class estimator vs. the peer-of-peers IV for λ and β . Right: GMM objective profiles — the correct estimator's minimum sits at the true $\beta = 5$, the naive one at ≈ 2 .

6. Extensions: weak instruments and a control/instrument sensitivity

Weak-instrument test. Regressing the endogenous norm S on the controls (x, Gx) and the excluded peers-of-peers characteristics G^2x , the partial F for G^2x is $\approx 1.3 \times 10^4$ (df 3, 24,778; partial $R^2 = 0.62$), far above the Stock–Yogo threshold: the peer-of-peers instruments are very strong here.

Sensitivity (control $\times$ instrument). The correct estimator combines two ingredients — controlling for direct-peer characteristics Gx and instrumenting the norm with G^2x . Table 2 and Figure 3 switch each on and off. The two ingredients play different roles. The peers-of-peers instrument G^2x is

Table 2. Only the combination (control + G^2x instrument) recovers both parameters.

Specification	$\hat{\lambda}$	$\hat{\beta}$
Naive (no Gx control, direct-peer IV)	0.924	1.97
Control only (Gx control, direct-peer IV)	0.768	2.04
Instrument only (no control, G^2x IV)	0.486	5.17
Correct (Gx control, G^2x IV)	0.298	4.98
True	0.300	5.00

what identifies the curvature: the two specifications that use it (instrument-only and correct) recover $\beta \approx 5$, the two that do not are stuck at $\beta \approx 2$. Recovering the peer effect λ additionally requires the contextual control — only the full combination reaches $\lambda = 0.30$. The control-only row is *weakly identified*: without a valid excluded instrument its $\hat{\beta}$ is unstable across seeds (ranging from ≈ 2 to ≈ 20), so it is reported only to isolate each ingredient’s role, not as a usable estimate. The naive and correct rows, and the first-stage $F \approx 1.3 \times 10^4$, are stable across seeds; the table fixes one seed (2026).

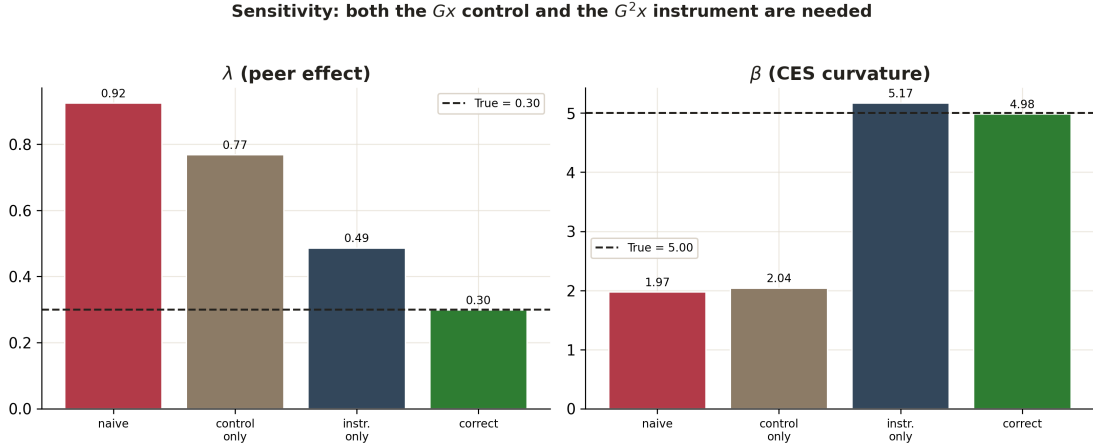


Figure 3. λ and β under the four specifications; only “correct” (green) matches the dashed true value in both panels.

7. Code modifications and reproducibility

All changes vs. the class code are marked in-line with [PP] comments.

- `DGP_pp.py`: [PP-1] heterogeneous school sizes (`generate_school_sizes`); [PP-2] contextual term $\phi'(Gx)$ added to the outcome fixed point.
- `Estimation_pp.py`: [PP] Gx added as a control; first stage augmented with G^2x ; instrument set $Z = (x, Gx, \hat{S}, \hat{D})$; plus the naive mode (class estimator) for comparison.

- `figures_pp.py`, `run_pp.py`: figures and a one-command driver.

Reproduce from the folder with `python run_pp.py` (fixed seed): it generates the data, runs both estimators, prints and saves the comparison table to `outputs_pp/`, and writes the figures to `figures_pp/`.

Extensions implemented. (i) a first-stage weak-instrument test for the peer-of-peers instruments (Section 6); (ii) a 2×2 sensitivity to the contextual control and the instrument choice; (iii) school-cluster-robust standard errors; (iv) heterogeneous schools. `run_pp.py` prints and saves all of these.